

14. The difference-equation method resulting from replacing $T^{(2)}(t, y)$ in Taylor's method of order two by $f(t + (h/2), y + (h/2)f(t, y))$ is known as :
- (A) Modified Euler method. (B) Milne Thomson method.
- (C) Midpoint method. (D) Adams Bashforth method.
15. Find approximation to the solution at $y(0.2)$ using Runge-Kutta method of order four for the IVP $y' = y - t^2 + 1, y(0) = 0.5$?
- (A) 1.21407. (B) 1.82929.
- (C) 0.82929. (D) 0.21407.

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2019

(CBCSS—UG)

Mathematics

MTS 5B 07—NUMERICAL ANALYSIS

(2019 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A*Answer at least eight questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 24.*

1. Show that $f(x) = x^3 + 4x^2 - 10 = 0$ has a root in $[1, 2]$.
2. Determine fixed points of the function $g(x) = x^2 - 2$.
3. Write the equation of Lagrange's interpolating polynomial through (x_0, y_0) and (x_1, y_1) .
4. State three point end point formula of differentiation.
5. Using Trapezoidal rule find $\int_0^2 x^2 dx$.
6. Show that $f(t, y) = t|y|$ satisfies a Lipschitz condition on the interval $D = \{(t, y) / 1 \leq t \leq 2 \text{ and } -3 \leq y \leq 4\}$.
7. Define a convex set.
8. For all $x \geq -1$ and any positive m show that $0 \leq (1+x)^m \leq e^{mx}$.
9. When is the initial value problem $\frac{dy}{dt} = f(t, y)$, $a \leq t \leq b$, $y(a) = \alpha$ well posed.
10. What is the degree of accuracy or precision of a quadrature formula ?

Turn over

11. Write Newton's Forward difference formula.
12. Set up Newton-Raphson formula for computing $\sqrt{N}$.

(8 × 3 = 24 marks)

Section B

Answer at least five questions.

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Find a root of $f(x) = x^3 - 3x - 5 = 0$ correct to 3 decimal places using Newton-Raphson method. Start with $x_0 = 3$.

14. Using Lagrange's interpolation formula find $y(10)$ if :

x	:	5	6	9	11
y	:	12	13	14	16

15. Using Newton's forward interpolation formula find the cubic polynomial for the data :

x	:	0	1	2	3
y	:	1	2	1	10

16. Approximate $\int_1^2 \frac{1}{x} dx$ using Simpson's $\frac{3}{8}$ th rule with step value $h = 0.25$

17. Using Second derivative midpoint formula approximate $f^{(1)}(1.3)$ if $f(x) = 3xe^x - \cos x$ with $h = 0.1$. Given :

x :	1.2	1.29	1.30	1.31	1.40
y :	11.59006	13.78176	14.04276	14.30741	16.86187

18. Use Euler's method to find approximate solution for the initial value problem $y' = 1 + \frac{y}{t}$, $1 \leq t \leq 2$, $y(1) = 2$ with $h = 0.25$.
19. Use Newton's Backward difference formula to construct interpolating polynomial of degree 1 if $f(-0.75) = -.07181250$, $f(-0.5) = -.02475000$, $f(-.25) = .33493750$, $f(0) = 1.10100000$.

(5 × 5 = 25 marks)

Section C

*Answer any **one** question.*

The question carries 11 marks.

20. Find by the method of Regula Falsi a root of the equation $x^3 + x^2 - 3x - 3 = 0$ lying between 1 and 2.
21. Use the Modified Euler method to approximate the solutions to the IVP $y' = \frac{1+t}{1+y}, 1 \leq t \leq 2,$
 $y(1) = 2$ with $h = 0.5$.

(1 × 11 = 11 marks)

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021**(CBCSS-UG)****Mathematics****MTS 5B 06—BASIC ANALYSIS****(2019 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 5B 06—BASIC ANALYSIS

(Multiple Choice Questions for SDE Candidates)

1. Which one of the following statements is true ?

- (A) If $n \in \mathbb{N}$, then there is an injection from $\mathbb{N}_n$ into $\mathbb{N}$.
- (B) If $n \in \mathbb{N}$, then there doesn't exist an injection from $\mathbb{N}_n$ into $\mathbb{N}$.
- (C) If $n \in \mathbb{N}$, then there is a bijection from $\mathbb{N}_n$ into $\mathbb{N}$.
- (D) None of the above options.

2. A set S is denumerable if _____.

- (A) There exists an injection from S into $\mathbb{N}$.
- (B) There exists a surjection from $\mathbb{N}$ onto S .
- (C) There exists a bijection from $\mathbb{N}$ onto S .
- (D) None of the above options.

3. The set A of all real numbers x such that $3x + 2 \leq 6$ is given by _____

(A) $A = \left\{ x \in \mathbb{R} : x \leq \frac{2}{3} \right\}.$

(B) $A = \left\{ x \in \mathbb{R} : x \geq \frac{4}{3} \right\}.$

(C) $A = \left\{ x \in \mathbb{R} : x \leq \frac{4}{3} \right\}.$

(D) None of the above options.

4. The set $S = \{x \in \mathbb{R} : x^2 + 2x > 3\}$ is _____.

(A) $S = \{x \in \mathbb{R} : x > 1\} \cup \{x \in \mathbb{R} : x < -3\}.$

(B) $S = \{x \in \mathbb{R} : x > 1\} \cup \{x \in \mathbb{R} : x < -2\}.$

(C) $S = \{x \in \mathbb{R} : x > 2\} \cup \{x \in \mathbb{R} : x < -3\}.$

(D) None of the above options.

5. Which one of the following is not a sequence ?

(A) $(1, 2, 1, 2, 1, \dots)$.

(B) $(1, 3, 5, 7, \dots, (2n-1), \dots)$.

(C) $\left(\frac{1}{2}, \frac{1}{6}, \frac{1}{12}, \frac{1}{18}, \frac{1}{24}, \dots\right)$.

(D) $(1, 2, 3, 4)$.

6. The first five terms of the sequence (x_n) , where $x_n = \frac{1}{n(n+1)}$, are _____.

(A) $1, 1, 1, 1, 1$.

(B) $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}, \frac{1}{7}, \frac{1}{8}$.

(C) $\frac{1}{2}, \frac{1}{6}, \frac{1}{12}, \frac{1}{18}, \frac{1}{24}, \dots$

(D) None of the above options.

7. n^{th} term of the sequence $\left(0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots\right)$ is _____.

(A) $\frac{n+1}{n}$.

(B) $\frac{n-1}{n}$.

(C) $\frac{n}{n-1}$.

(D) $\frac{n}{n+1}$.

8. The sequence $\{\sqrt{n}\}$ _____.

(A) Converges to 0.

(B) Converges to 1.

(C) Converges to $\sqrt{2}$.

(D) Diverges.

9. The Fibonacci sequence is given by _____.

- (A) $(1, 2, 2, 3, 5, 8, 13, 21, 34, 55, \dots)$.
- (B) $(1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \dots)$.
- (C) $(1, 1, 2, 4, 5, 8, 13, 21, 34, 55, \dots)$.
- (D) $(1, 2, 3, 4, 5, 8, 13, 21, 34, 55, \dots)$.

10. If X and Y are the sequences $X = (2, 4, 6, \dots, 2n, \dots)$, $Y = \left(\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots\right)$, then $X - Y =$ _____.

- (A) $\left(\frac{1}{2}, \frac{7}{2}, \frac{17}{3}, \dots, \frac{2n^2 - 1}{n}, \dots\right)$.
- (B) $\left(\frac{1}{1}, \frac{7}{2}, \frac{17}{3}, \dots, \frac{2n^2 - 1}{n}, \dots\right)$.
- (C) $\left(\frac{1}{1}, \frac{3}{2}, \frac{7}{3}, \dots, \frac{2n^2 - 1}{n}, \dots\right)$.
- (D) None of the above options.

11. $\lim\left(\frac{2n}{n^2 + 1}\right) =$ _____.

- (A) 0.
- (B) 1.
- (C) 2.
- (D) 3.

12. $\lim\left(\frac{\sin n}{n}\right) =$ _____.

- (A) 1.
- (B) 2.
- (C) 3.
- (D) None of the above options.

13. The sequence $S = (\sin n)$ _____.
- (A) Is divergent. (B) Converges to 0.
(C) Converges to 1. (D) None of the above options.
14. The sequence $\left(\sin \frac{n\pi}{4}\right)$
- (A) Is divergent. (B) Converges to 0.
(C) Converges to 1. (D) None of the above options.
15. $\left(1 + \frac{1}{2!} + \dots + \frac{1}{n!}\right)$ _____.
- (A) Is a Cauchy sequence. (B) Not be Cauchy sequence.
(C) Not convergent. (D) None of the above options.
16. Which one of the following is a false statement ?
- (A) There are bounded sequences that are not Cauchy.
(B) $\left((-1)^n\right)$ is not a Cauchy sequence.
(C) If (x_n) is a Cauchy sequence such that x_n is an integer for all $n \in \mathbb{N}$, then (x_n) is ultimately constant.
(D) If (x_n) is a Cauchy sequence such that x_n is an integer for all $n \in \mathbb{N}$, then (x_n) is a constant sequence.
17. Let $F \subseteq \mathbb{R}$ be such that if $X = (x_n)$ is any convergent sequence of elements in F , then $\lim X$ belongs to F , then _____.
- (A) F is a closed set. (B) F is an open set.
(C) F is neither closed nor open. (D) F is both open and closed.
18. The Cantor set _____.
- (A) Has only finite number of points. (B) Has countable number of points.
(C) Is denumerable. (D) Is uncountable.